

MD5 (MESSAGE DIGEST ALGORITHM 5)

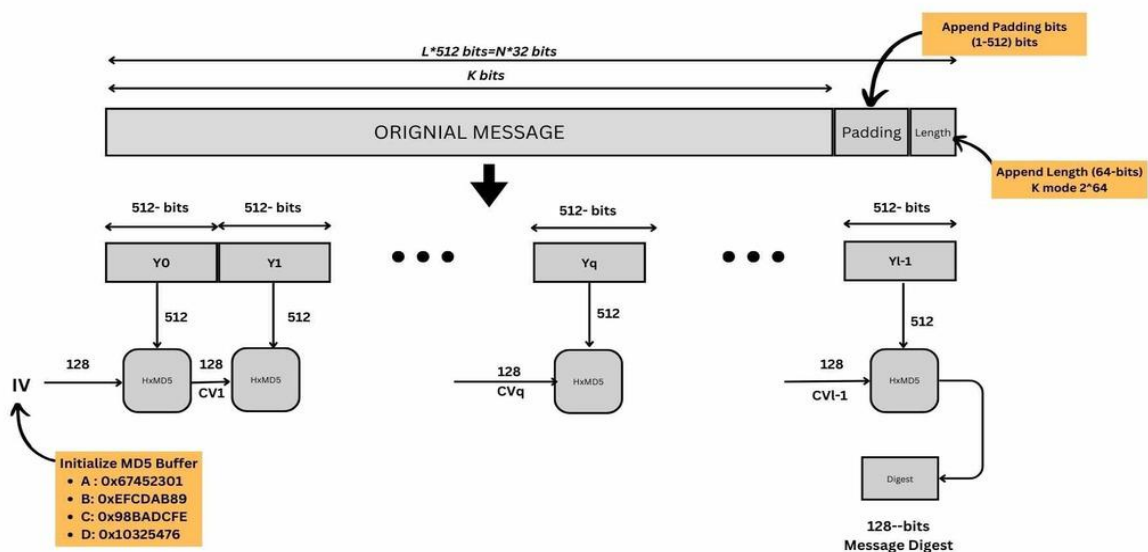
1.INTRODUCTION:

MD5 (Message Digest Algorithm 5) is a cryptographic hash function that takes input data of any length and converts it into a fixed-length output of 128 bits (16 bytes). It was developed in 1991 by Ronald Rivest as an improved version of MD4, aiming to provide better security. Although MD5 was widely used for security purposes like password hashing and data integrity verification, it is now considered cryptographically broken due to vulnerabilities such as collision attacks. However, it is still used in some non-cryptographic applications like checksums and file integrity verification.

2.CONCEPT:

- MD5 stands for Message Digest Algorithm 5
- Developed in 1991 by Ronald Rivest
- It is a cryptographic hash function
- Accepts input of any length
- Produces a fixed output of 128 bits (16 bytes)
- Output is usually shown as 32-character hexadecimal value

3.BLOCK DIAGRAM:



4.IMPLEMENTATION:

1. Input Message
 - Take message of any length
2. Padding
 - Add 1 bit (1) followed by 0s
 - Make message length $\equiv 448 \bmod 512$
3. Append Length
 - Add 64-bit representation of original message length
 - Now total size becomes multiple of 512 bits
4. Initialize Buffers
 - Initialize 4 registers:
 - $A = 0x67452301$
 - $B = 0xEFCDAB89$
 - $C = 0x98BADCFE$
 - $D = 0x10325476$
5. Process Each 512-bit Block
 - Divide into blocks: $Y_0, Y_1, \dots Y_n$
 - Each block undergoes 4 rounds (16 operations each = 64 steps)
6. Use Auxiliary Functions
 - Four nonlinear functions used:
 - $F(B,C,D) = (B \text{ AND } C) \text{ OR } (\text{NOT } B \text{ AND } D)$
 - $G(B,C,D) = (B \text{ AND } D) \text{ OR } (C \text{ AND NOT } D)$
 - $H(B,C,D) = B \text{ XOR } C \text{ XOR } D$

- $I(B,C,D) = C \text{ XOR } (B \text{ OR NOT } D)$

7. Update Values

- A, B, C, D values are updated after each round
- Output of one block → input to next (chaining)

8. Final Output

- Combine A, B, C, D → produce 128-bit hash (message digest).

5. REAL TIME EXPLANATION:

1. API Request Signing (Basic Systems)

- Some simple systems use MD5 to create a signature for API requests
- Helps check if request is modified or not

2. Email Systems (Message Integrity)

- MD5 used to ensure email content is not altered
- Verifies message integrity during transfer

3. Blockchain / Early Crypto Usage

- Early systems used MD5-like hashing for data linking
Now replaced with stronger hashes

4. File Naming / Unique ID Generation

- MD5 used to generate unique IDs for files
- Example: storing files with hashed names

5. Content Management Systems (CMS)

- Used to detect changes in website content
- Helps in version tracking

6. Gaming Systems

- Used to verify game files integrity
- Prevents modified/cheat files

7. Cloud Storage Systems

- Used to verify file upload/download correctness
- Ensures no corruption in cloud transfer